Async Lab: The Green AI Trade-Off

📌 1. Objective

In this lab, you will act as a Lead MLOps Engineer tasked with optimizing a model for a production environment where cloud computing budgets and carbon emissions are strictly monitored. You must determine if utilizing the full dataset and training for extended epochs provides a justifiable Return on Investment (ROI) in terms of accuracy vs. carbon emitted.

🛠️ 2. Architectural Requirements

1. **Dataset:** Choose any Computer Vision dataset (e.g, Fashion-MNIST, CIFAR-10, SVHN). *Hint: tf.keras.datasets has many built-in options.*
2. **Model Architecture:** You can use a CNN (Conv2D layers). You must build a Deep Multi-Layer Perceptron (MLP).
   - **Requirement:** 1 Flatten() layer followed by **at least 5** Dense() hidden layers, ending with your output classification layer.
3. **Tracking:** You must wrap your training loops using the codecarbon library's EmissionsTracker.

🧪 3. The Experiment Matrix

You must write a Python script that trains 4 separate models from scratch using the following configurations:

- **Experiment A:** 50% of the Training Data, trained for 50 Epochs.
- **Experiment B:** 50% of the Training Data, trained for 100 Epochs.
- **Experiment C:** 100% of the Training Data, trained for 50 Epochs.
- **Experiment D:** 100% of the Training Data, trained for 100 Epochs.

(Hint: To get 50% of the data, you can use slicing, e.g., X_train_half = X_train[:len(X_train)//2])

📊 4. Deliverables & Visualization

Your final Python notebook/script must automatically generate a plot (or subplots) that clearly visualizes the trade-offs. Your plot should display:

- The 4 Experiments on the X-axis (or grouped logically).
- **Test Accuracy (%)** for each experiment.
- **CO2 Emissions (grams)** for each experiment.
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🏆 5. The Conclusion (README.md)

You must write a short conclusion identifying the "Winning Configuration." Answer the following questions:

1. Did doubling the dataset size double the accuracy? Did it double the emissions?
2. Did doubling the epochs from 50 to 100 provide a meaningful accuracy boost, or did it just waste electricity?
3. Which of the 4 models would you deploy to production and why?

🚀 6. GitHub Submission Protocol

You must submit this lab via the Open Source PR workflow we established previously.

1. Open your Colab Notebook and execute your 4 experiments.
2. Save your visualization as an image (e.g., tradeoff_plot.png).
3. Create a new folder in your repository via the terminal:
   !mkdir -p week-10-12-async-lab
4. Move your code, the emissions.csv file, your plot image, and a README.md (containing your conclusion) into this folder.
5. Add, Commit, and Push to your GitHub account:
   Bash
   ```
   !git add week-10-12-async-lab/
   !git commit -m "feat: completed Deep MLP carbon vs data size experiments"
   !git push origin main
   ```